TECHNICAL EVALUATION REPORT

The Physics of the Instant: Analysis from Limits, Relativity, and Renormalization

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1 Summary of the submission

The paper is an essay on conceptual aspects of element involved in physical theories. The paper argues that the *dynamical completeness* of the instantaneous state of a physical system (the property that the state at a temporal point suffices to fix future evolution) is a physical assumption rather than a consequence of how that state is defined, and that established physics supplies no basis for it and positive reasons to doubt it.

Three independent lines are assembled: limits, quantum-gravitational resolution bounds, and renormalization/QFT localization; and then draws consequences for Markov descriptions, Chapman-Kolmogorov (CK) composition, and one timing formulation of the quantum measurement problem. Section §2 observes that the ε–δ definition of a limit characterises a function on punctured neighbourhoods of a point and asserts nothing about the point itself, so that the construction delivers an element of the state space and nothing about its dynamical sufficiency. Section §3 assembles the Mead–Garay and 't Hooft minimum-length arguments and the Salecker–Wigner clock bound, the latter combined with the requirement that the clock not collapse, to conclude that the limiting process $\delta t \to 0$ cannot be physically implemented. Section §4 observes that products of field operators at coincident points are singular, that physical content in quantum field theory is extracted at finite resolution, and that the algebraic formulation assigns observables to extended regions. Section §5 applies the conclusion to what the paper names the *collapse-timing formulation* of the measurement problem. Section §6 draws the consequence for memoryless descriptions and states the paper's one testable prediction. Section §7 states the originality claim: not any of the ingredients, but the use to which they are jointly put.

The paper is conceptual and inferential and present arguments about fundamental aspects. Doctoral rigour therefore turns on the validity of the bridges it builds (between mathematical definition, operational limitation, field localisation, Markov sufficiency and Chapman–Kolmogorov (CK) composition). The paper is accordingly judged here on four axes: whether each argument is valid and correctly scoped; whether the physics it invokes is correctly stated; whether the testable prediction is both true and discriminating; and whether the joint assembly is original.

2 Overall assessment

The paper shows a strong argumentative tone that tends to persuade the reader about the arguments it tries to establish. The paper is correct about the fact that the epsilon-delta definition of a temporal limit does not itself imply that the limiting state is dynamically sufficient. In fact this is not an issue for since as recognized later in the text: *"What the construction cannot supply is that this element determines the future, because sufficiency is a property of the dynamics, the existence of a law that fixes later states from the present one alone, and not a property of the limit."* More over one should bear in mind that the notion of state in most theories is a primitive concept; an in general is not defined via limits from the mathematical point of view. So, for example the claim: *"The state space may carry whatever structure the theory requires, a phase space or a Hilbert space, and the limit is taken in that space; the limit is then an element of the space with all of the structure that membership confers."* has not a firm ground. The argument here is at least vague since the problem is not clearly set out.

Beyond that, every quantitative statement seems to be correct; in particular: the Salecker–Wigner mass condition, its combination with the no-collapse and readability conditions, the resulting bound $\delta t \gtrsim (T t_P^2)^{1/3}$, its corollary $\delta t \gtrsim t_P$ at $T \sim \delta t$, and the numerical values of t_P and ℓ_P are all correct as printed. The treatment of renormalization is accurate on regulator removal, on

the theory-dependence of the interacting continuum limit, and on ϕ^4_4 triviality.

Also, the paper's testable prediction, as stated in §6, seems to be correct but do not discard that there could exist process with memory not arising from the the limitations associated to finite resolution descriptions.

Against this, two material defects must be recorded. Both are remediable, and neither touches the verified physics of §3.

First, the paper conflates Chapman–Kolmogorov composition with memorylessness and with dynamical completeness (Finding 6). Markov implies CK for the associated transition kernels; the converse fails, and has been known to fail since Feller's 1959 example of a non-Markov process whose two-time kernels compose. The abstract, §6 and §7 treat the two as interchangeable. The damage is narrower than it first appears (the inferences §6 actually draws run in the valid direction of the implication) but the identification is stated repeatedly and must be corrected, and its root cause is that "dynamical completeness" is never given a mathematical definition.

Second, the unconditional claim of §2 is false as written (Finding 2). Section §2 asserts that for a process observed through a temporal window the defect "vanishes only as $\delta \to 0$". That fails whenever the observation is a non-lumpable function of the state (which is precisely the case treated by the two authors the sentence cites. The correctly conditioned statement is the one §6 already gives; §2 must import its hypothesis.

Three further qualifications are recorded below: an overstatement of what the algebraic formulation supports, which omits the one axiom that bears directly on initial-value structure (Finding 5); a conflation of the renormalization scale with a physical resolution (Finding 4); and a contradiction between §7's claim to remain within confirmed physics and §3's own statement that it extrapolates beyond it.

3 Section-by-section analysis

3.1 The argument from limits (§2)

Finding 1 (Valid observation, but the apparatus is not what carries it)**.** The conclusion of §2 is that dynamical sufficiency is a property of the dynamics rather than of the construction of the state. That is true, and it follows immediately from what dynamical sufficiency means. The ε–δ machinery, the punctured neighbourhood and the $\sin(x)/x$ example establish that a limit is independent of any value separately assigned at the point (a correct and standard observation, but not the premise the conclusion requires).

The tension is visible in the section's own structure. If theories introduce point-indexed states *directly*, as its first sentence concedes, the state is not constructed as a limit and the question of inheritance does not arise. If instead the state is read operationally as a limit of interval averages, the identification of a measurement model with the definition of the state is itself a substantive assumption, which the section does not defend. Either way the conclusion stands, by a shorter route. The section is the longest of the three and yields the least.

Finding 2 (The unconditional window claim is false, and its own authorities refute it)**.** The final paragraph of §2 states that for a process observed through a temporal window of finite width, "the observed transition family fails to compose through intermediate times for every $\delta > 0$, with a defect that vanishes only as $\delta \to 0$", citing Burke and Rosenblatt (1958) and Kemeny and Snell (1960).

The second half of that sentence is false without further hypotheses. Those two references concern a *function of the state* (the lumping of a state space) and what they establish is that a

lumping of a Markov chain is generically *not* Markov. If the observation is such a lumping and the partition is not lumpable, then the defect does *not* vanish as $\delta \to 0$: it converges to the defect of the lumped two-time family, which is positive. The authorities cited therefore refute the sentence they are cited for.

Two further slippages compound this. The manuscript's observable is a functional of the *path segment* within the window, which is a different operation from a lumping of the state: the sentence says so itself, "the observed quantity is a function of the path segment within the window", and then cites the lumping literature. And neither reference addresses the behaviour of the defect *under refinement*, which is the claim that carries section §6.

The repair is available in the manuscript itself. Section §6 states the conditioned version ("for a system that is Markov on the observed state space, the composition defect of the window-averaged family must fall to zero as the window is refined") and that version is correct. Section §2 must import that hypothesis, and must say which of the two operations (temporal windowing, state-space lumping, or their composition) is in force at each step. **[M]**

3.2 The arguments from relativity (§3)

The scoping here is exemplary and should be preserved. The section states at the outset that the bounds "are obtained by extrapolating both theories toward a regime where neither is independently known to remain complete", that they "are not theorems of a completed theory of quantum gravity", and that a minimum measurable resolution is to be distinguished throughout from spacetime discreteness and from the failure of point-indexed mathematical states.

The Mead–Garay and 't Hooft routes. Correctly described, and correctly characterised as semiclassical evidence for an operational limit rather than as theorems. The observation that 't Hooft's route "depends only on the particle kinematics and the Einstein equations" and therefore "rests on fewer assumptions" is a fair comparison.

The Salecker–Wigner route. The minimum-mass condition $M \gtrsim \hbar T/(c^2 \delta t^2)$, its combination with $r_s = 2GM/c^2 \lesssim \ell_{\text{clock}} \lesssim c\,\delta t$, and the resulting Note that the bound is $\delta t^3 \gtrsim T\,t_P^2$, in units of time throughout; a rendering in terms of ℓ_P^2 is dimensionally inconsistent and should not be adopted. The corollary $\delta t \gtrsim t_P$ at $T \sim \delta t$ is correct. The dispute over the T-dependence is acknowledged with both sides cited, and the paper confines itself to the Planck-scale corollary, which the dispute does not touch. The statement that this route "is independent of the Mead and 't Hooft analyses in that it constrains the clock rather than the probed region, not in that it dispenses with gravitation" is correct.

Finding 3 (An unresolved tension over time–energy uncertainty)**.** The section infers a temporal bound from the spatial results — "probing temporal intervals shorter than t_P requires concentrating energy at the same densities that trigger gravitational self-collapse" — flagged honestly as the paper's own inference. Two paragraphs later it states that the bounds "do not rely on the time–energy uncertainty relation".

As written the inference reads as an application of the relation being disclaimed. The disclaimer is worth keeping; the inference then needs its own licence, and one is available at the cost of a sentence: a signal localised in time to δt is localised in space to $c\,\delta t$, and localisation to $\delta x = c\,\delta t$ requires momentum $\hbar/\delta x$ and hence energy $\hbar/\delta t$ by the *position–momentum* relation, which is uncontroversial. This is presumably what "the energy scales are identical, related by the factor c" is meant to convey. **[R]**

3.3 The argument from renormalization (§4)

Several statements deserve specific endorsement. That "where a continuum limit exists, it is obtained by removing the regulator from renormalized quantities, not by evaluating the bare theory at a point, and whether an interacting continuum limit exists at all is theory-dependent" is correct, and forecloses the common overstatement that all quantities at a point are infinite. The citation of ϕ^4_4 triviality to Aizenman and Duminil-Copin is apt and correctly restricted to its lattice/critical setting. The lattice discussion (finiteness at fixed spacing, with renormalization entering in relating descriptions across scales) is correct.

Finding 4 (The renormalization scale is not a physical resolution)**.** The section writes that "physical predictions are defined at finite resolution" and that "a theory at energy scale μ is an effective description whose parameters run with the scale". These sentences run together four distinct quantities that must be separated: the renormalization scale μ; the ultraviolet regulator; the physical validity scale Λ of an effective description; and experimental resolution.

The point that matters is that exact physical observables are *independent* of μ (that independence is the content of the renormalization-group equation) so the running of renormalized parameters with μ is not evidence that physics is defined only at finite resolution. The running of α_s is measured, as the paper says, but what is measured is the value of a scheme-dependent coupling at a chosen scale, not a limit on resolution.

The section's conclusion nonetheless survives, by a different route: an effective field theory has a physical validity scale Λ above which new degrees of freedom enter, and predictions are organised in powers of E/Λ. That is a genuine statement about finite resolution. The manuscript should make the argument through Λ and drop the appeal to μ, which does not support it. **[R]**

Finding 5 (The algebraic formulation is invoked without its time-slice axiom)**.** The section invokes the Haag–Kastler assignment of observable algebras to open regions, and then (correctly, and to its credit) adds that those results "concern the localization of observables; they do not by themselves bear on the composition of the dynamics, which in that framework is carried by the automorphism group". The concession is right but incomplete, because it omits the one structural feature of the algebraic framework that bears directly on the paper's thesis.

The time-slice axiom (primitive causality) states that the algebra associated with an arbitrarily thin neighbourhood of a Cauchy surface already generates the algebra of the whole spacetime. That is an initial-value structure, stated inside the framework the paper is citing. A reader who knows the axiom and reaches §4 as written will conclude that algebraic quantum field theory tells against the paper rather than for it.

The honest reading is more interesting than either extreme, and the paper should give it. The time-slice axiom delivers determination from a slab of *nonzero* thickness, however small, while the sharp surface itself carries no algebra. The framework therefore locates the content in thin extended regions and not at a sharp time (which is closer to the manuscript's own thesis than the current phrasing manages to convey, but which cannot be had without confronting the axiom directly). As it stands the section takes half of the available support and exposes itself to the other half. **[R]**

One attribution to tighten. The Wightman reference is bibliographically correct: *Ann. Inst. Henri Poincaré A* **1**(4) (1964), 403–420, which I confirmed at source. The substantive attribution (that fields defined pointwise "lead to contradictions under natural assumptions") is stated too loosely to be checked: which contradiction, under which hypotheses? The paper should state the theorem. **[R]**

3.4 The collapse-timing formulation (§5)

The section does what it says. It names the formulation it addresses, identifies that demand as an instance of the completeness assumption, and argues that without the assumption the question has no well-defined terms. The claim is dissolution of a *formulation*, not solution of the measurement problem, and the enumeration of what is not addressed (Born rule, definite-outcome problem, preferred-basis problem) is correct and complete for its purpose.

One qualification of scope. That every stage of a measurement takes nonzero time establishes that measurement interactions and record formation have finite duration, and that no unique operational collapse instant is singled out. It does not establish that no theoretical jump time can exist. The section's conclusion should be stated in the operationally modest form: there is no operationally privileged instant, rather than there is no instant. The statements drawn from the stochastic–quantum correspondence are attributed and are scoped to finite configuration spaces and unistochastic transition matrices, which is correct; they should continue to be presented as conclusions within that framework rather than as consequences of standard quantum theory. **[R]**

3.5 Consequences and the testable prediction (§6)

This is where the paper earns or loses its claim to a testable hypothesis, which is the substantive standard applied to theoretical work.

The prediction is that for a system Markov on the observed state space the composition defect of the window-averaged family must fall to zero as the window is refined, whereas a defect that persists under refinement is memory intrinsic to the chosen state space. The paper supplies its own principal caution: classical hidden variables also produce a resolution-independent defect, so the criterion does not separate classical memory from quantum indivisibility. That caution is correct and necessary, and stating it is what makes the prediction defensible.

Finding 6 (Chapman–Kolmogorov composition is not memorylessness)**.** The manuscript treats CK composition, the Markov property and dynamical completeness as one condition. The abstract lists "Markov dynamics and the Chapman-Kolmogorov composition of transition kernels" as the descriptions that inherit the completeness assumption; §6 calls them "the principal cases"; §7 lists "the Markov property, the composition of transition kernels through intermediate times" in a single series.

They are not the same condition. Markov implies CK for the associated two-time kernels; the converse fails. The standard witness is W. Feller, "Non-Markovian processes with the semigroup property", *Ann. Math. Statist.* **30** (1959), 1252–1253, which exhibits a non-Markov process whose transition probabilities satisfy the Chapman–Kolmogorov equation. The Markov property constrains the whole hierarchy of finite-dimensional distributions; CK constrains only the two-time conditionals. A vanishing CK defect therefore does not establish memorylessness, and *a fortiori* does not establish dynamical completeness.

A correction required that about the framing, and it is not cosmetic: the paper's thesis is about *completeness*, its evidence is about *CK composition*, and the step between them is the one the paper never licenses. The root cause is that "dynamical completeness" is nowhere given a mathematical definition. It should be defined separately for the two settings: well-posed evolution from admissible initial data in the deterministic case, and $\mathbb{P}(X_t \in A \mid \mathcal{F}_s) = \mathbb{P}(X_t \in A \mid X_s)$ for $s < t$ in the stochastic case — and the hierarchy Markov $\Rightarrow$ CK, with the converse failing, should be stated where the two first appear together. **[M]**

3.6 Discussion (§7)

The originality claim is correctly limited: "what is new is not any of these results but the use to which they are jointly put". The concluding formulation: that the arguments "remove the basis for treating the completeness of the instantaneous state as given", is the weaker and correct form.

One sentence contradicts the body. Section §7 states that the arguments do their work "without leaving the domain of confirmed physics". Section §3 states that its bounds are obtained "by extrapolating both theories toward a regime where neither is independently known to remain complete". Both cannot stand. §3 has it right, and §7 should be brought into line: the semiclassical status of the Planck-scale reasoning must be preserved wherever the conclusion is restated. [**R**]

4 Originality: what is new and what is standard

The paper's own. The joint assembly, and specifically the observation that three independent bodies of established physics converge on the same conclusion about the temporal point from three unrelated directions — construction, measurement, and formalism. The naming and isolation of the *collapse-timing formulation*, with the explicit enumeration of what it does not address. The resolution-scaling criterion of §6, which converts a conceptual thesis into a measurable quantity.

Level. The originality is moderate as a foundational synthesis and low as new mathematics or new physics, and the paper's own framing in §7 says as much.

5 Priority in the literature

The ingredients are standard and are not claimed. I found no prior work assembling these three lines against the dynamical completeness of the instantaneous state, and none proposing the defect-scaling criterion of §6. The general territory is occupied (the Planck-scale measurement-limit literature is reviewed by Garay, whom the paper cites) but the specific conjunction and the specific criterion did not surface.

Two positioning items should be added. [**R**] The philosophical literature on the status of the present is the nearest neighbouring discussion to §2 and §5 and is not engaged. And §6's caution about separating classical from quantum memory points at the quantitative divisibility-failure measures for open quantum systems and at the process-tensor treatment of classical versus quantum non-Markovianity; the paper cites these as "the natural starting point" but does not say what the criterion drawn from them would be, leaving its most important open question unspecified.

6 Specific criticisms

Items are marked [**M**] where they bear on the correctness or the substance of a claim, and [**R**] where they concern presentation, attribution or scope.

[M] **Chapman–Kolmogorov composition is identified with memorylessness and with dynamical completeness**
(Finding 6). Markov ⇒ CK; the converse fails. Define dynamical completeness mathematically for the deterministic and stochastic settings, and state the hierarchy with its counterexample where the two notions first appear together.

[M] The unconditional window claim of §2 is false as written
(Finding 2), and the authorities cited for it establish its negation in the case they treat. A counterexample is recorded in the verification section. The conditioned statement of §6 is correct and should replace it.

[R] The renormalization scale is treated as a physical resolution
(Finding 4). Separate μ, regulator, EFT validity scale and experimental resolution; the conclusion survives through the validity scale.

[R] The algebraic formulation is invoked without the time-slice axiom
(Finding 5), which is the one feature of that framework bearing directly on initial-value structure.

[R] §7 contradicts §3
on whether the argument remains within confirmed physics.

[R] §2 is disproportionate to its yield
(Finding 1).

[R] Unresolved tension over time–energy uncertainty in §3
(Finding 3).

[R] The Wightman attribution is too loose to be checked.

[R] §5 should be stated in the operationally modest form:
no operationally privileged instant, rather than no instant.

[R] Positioning gaps:
the philosophical literature on the status of the present; and the criterion that would separate classical from quantum memory.

7 Questions to the author

1. Define dynamical completeness mathematically, separately for deterministic and for stochastic theories, and say which of the two your thesis concerns at each point of the argument.

2. How do Feller-type processes — non-Markov, but with Chapman–Kolmogorov-composing two-time kernels — bear on your thesis? Your evidence is about CK composition and your conclusion is about completeness; state the step that connects them.

3. Restate the window claim of §2 with the hypothesis that §6 already supplies, and account for the case of a Markov process observed through a non-lumpable function of its state, whose CK defect does not vanish under refinement.

4. Supply a proof, or a precise reference, for the refinement limit under the conditioned hypothesis, with the conditions and if possible a rate. Note that Burke–Rosenblatt and Kemeny–Snell concern lumpings of the state space rather than functionals of a path segment, and address the existence of a defect rather than its behaviour under refinement.

5. Incorporate the two strengthenings of Finding **??**, and specify the sampling floor a protocol must report.

6. Give the licence for the inference from spatial to temporal bounds without invoking the time–energy uncertainty relation you disclaim.
7. How does the time-slice axiom affect your use of the algebraic formulation, and why should the singularity of point-localised fields bear on the completeness of Cauchy data?
8. Distinguish the renormalization scale, the regulator, the effective-theory validity scale and experimental resolution, and say which of them your argument requires.
9. Reconcile §7's claim to remain within confirmed physics with §3's statement that it extrapolates beyond it.
10. What empirical outcome would falsify the foundational thesis, as distinct from falsifying a chosen observation model?

8 Technical conclusion

Every quantitative statement in this manuscript seems to be correct. The paper's testable prediction, in the conditioned form §6 gives it, is correct with some caveats we mentioned previously.

Two material defects stand between the manuscript and the thesis it states. Its evidence concerns Chapman–Kolmogorov composition while its conclusion concerns dynamical completeness, and the two are identified rather than connected; the root cause is that completeness is never defined. And the unconditional form in which §2 states the window claim is false, by a counterexample recorded here, in exactly the case the authorities it cites are about. Both are remediable without new physics: the first by a definition and a statement of the Markov–CK hierarchy, the second by importing into §2 the hypothesis that §6 already carries.

Three further corrections concern the scope of the supporting evidence: the renormalization scale does not bear the weight placed on it, though the conclusion survives through the effective-theory validity scale; the algebraic formulation cannot be invoked without its time-slice axiom; and the claim to remain within confirmed physics contradicts the manuscript's own account of what it does.